STATISTICS 652

LECTURE 13 - 2026

LARGE SAMPLE THEORY FOR M.L.E. (MULTIPARAMETER CASE)

Suppose $\underset{\sim}{X}_1, \ldots, \underset{\sim}{X}_n \in \mathbb{R}^k$ represents a random sample of $\underset{\sim}{X} \sim F$ with $F \in \mathcal{F} = \{F_{\underset{\sim}{\theta}} : \underset{\sim}{\theta} \in \Theta\}$, $\Theta \subset \mathbb{R}^q$, Θ open, $\text{vol}[\Theta] > 0$.

Let $\ell(\underset{\sim}{\theta}; \underset{\sim}{x}) = \log f(\underset{\sim}{x}; \underset{\sim}{\theta})$ where $f(\underset{\sim}{x}; \underset{\sim}{\theta})$ is the density or probability mass function associated with $F_{\underset{\sim}{\theta}}(\underset{\sim}{x})$.

UNDER SOME TECHNICAL REGULARITY CONDITION OF $f(\underset{\sim}{x}; \underset{\sim}{\theta})$

LEMMA 13.1. For $\underset{\sim}{\theta}_0 \in \Theta$, let $\gamma_0(\underset{\sim}{\theta}) = E_{\underset{\sim}{\theta}_0}[\ell(\theta; X)]$, then $\gamma_0(\underset{\sim}{\theta}_0) \geq \gamma_0(\underset{\sim}{\theta})$ with equality IFF $\underset{\sim}{\theta} = \underset{\sim}{\theta}_0$.

PROOF: (ANALOGOUS TO THE ONE PARAMETER CASE)

$$\gamma_0(\underset{\sim}{\theta}) - \gamma_0(\underset{\sim}{\theta}_0) = E_{\underset{\sim}{\theta}_0}\left[\log\{f(\underset{\sim}{X}; \underset{\sim}{\theta}) / f(\underset{\sim}{X}; \underset{\sim}{\theta}_0)\}\right]$$

$$\leq \log E_{\underset{\sim}{\theta}_0}\left[f(\underset{\sim}{X}; \underset{\sim}{\theta}) / f(\underset{\sim}{X}; \underset{\sim}{\theta}_0)\right] \quad \text{SINCE } \log \text{ IS CONCAVE SO JENSEN'S INEQUALITY APPLIES}$$

$$= \log 1 = 0 \quad . \square$$

NOTE: LEMMA 13.1 STATES THAT THE MLE IS FISHER-CONSISTENT.

DEFINITION 13.2. AN ESTIMATOR $\hat{\underset{\sim}{\theta}} = \underset{\sim}{T}(F_n)$, i.e. $\hat{\theta}$ can be expressed AS A FUNCTIONAL OF THE EMPIRICAL DISTRIBUTION, IS SAID TO BE FISHER CONSISTENT IFF $\underset{\sim}{T}(F_{\underset{\sim}{\theta}}) = \underset{\sim}{\theta}$, all $\underset{\sim}{\theta} \in (H)$, FOR $\underset{\sim}{\theta}$

EXAMPLE: $\bar{\underset{\sim}{X}} = E_{F_n}(\underset{\sim}{X})$ and $\mu = E_F(\underset{\sim}{X})$, AND SO THE SAMPLE MEAN IS FISHER CONSISTENT FOR THE POPULATION MEAN, IF IT EXISTS,

NOW, M.L.E., $\hat{\underset{\sim}{\theta}}_n = \underset{\underset{\sim}{\theta} \in (H)}{\text{ARG SUP}} \frac{1}{n} \sum_{i=1}^{n} l(\underset{\sim}{\theta}; \hat{\underset{\sim}{X}}_i)$

$= T(F_n) = \underset{\underset{\sim}{\theta} \in (H)}{\text{ARG SUP}} E_{F_n}(l(\underset{\sim}{\theta}; X))$

AND LEMMA 15.1 STATES $\underset{\sim}{\theta} = T(F_{\underset{\sim}{\theta}})$, HENCE FISHER CONSISTENT FOR $\underset{\sim}{\theta}$.

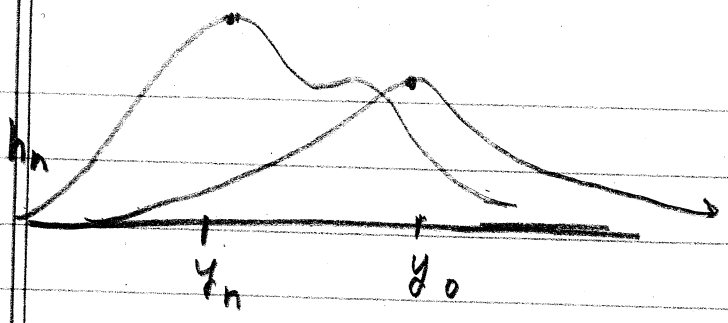
THEOREM 13.3a: $\hat{\underset{\sim}{\theta}}_n \to \underset{\sim}{\theta}$ almost surely.

PROOF: (SKETCH) LET $\varphi_n(\underset{\sim}{\theta}) = \frac{1}{n} \sum_{i=1}^{n} l(\underset{\sim}{\theta}; \hat{\underset{\sim}{X}}_i)$, SO

BY S.L.L.N. $\varphi_n(\underset{\sim}{\theta}) \xrightarrow[\text{A.S.}]{} \varphi_0(\theta) = E_{\underset{\sim}{\theta}_0}[l(\underset{\sim}{\theta}; \hat{\underset{\sim}{X}}_i)]$ POINTWISE, i.e. FIXED $\underset{\sim}{\theta}$

NOTE: $E_{\underset{\sim}{\theta}_0}$ MEANS EXPECTATION UNDER $F_{\underset{\sim}{\theta}_0}$.

HW: SHOW IF $h_n : \mathbb{R}^p \to \mathbb{R}$ and $h : \mathbb{R}^p \to \mathbb{R}$ are continuous and bounded above with $h(\underset{\sim}{y})$ having a unique maximum at $\underset{\sim}{y} = \underset{\sim}{y}_0$, and $h_n \to h$ pointwise, then



$\underset{\sim}{y}_n = \arg\sup_{\underset{\sim}{y}} h_n(\underset{\sim}{y}) \to \underset{\sim}{y}_0$.

NOTE: ADDITIONAL ADVANCED ARGUMENTS ARE NEEDED TO CONCLUDE THAT $\hat{\underset{\sim}{\theta}}_n \to \underset{\sim}{\theta}$ AS n, SINCE THE SET OF PROBABILITY ONE FOR WHICH $\psi_n(\underset{\sim}{\theta}_1) \to \psi_0(\underset{\sim}{\theta}_1)$ MAY DIFFER FROM THE SET OF PROBABILITY ONE OR WHICH $\psi_n(\underset{\sim}{\theta}_2) \to \psi_0(\underset{\sim}{\theta}_2)$.

THEOREM 13.3b: $\sqrt{n}(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}) \to N_2(\underset{\sim}{0}, I(\underset{\sim}{\theta}; \underset{\sim}{X})^{-1})$. (ASYMPTOTIC FISHER EFFICIENT)

PROOF. (SKETCH: ANALOGOUS TO ONE PARAMETER CASE)

Let $\underset{\sim}{\psi}(\underset{\sim}{X}; \underset{\sim}{\theta}) = \nabla \ell(\underset{\sim}{\theta}; \underset{\sim}{X})$ (2×1) gradient WRT $\underset{\sim}{\theta} \in \mathbb{R}^2$

RECALL $E_{\underset{\sim}{\theta}}[\underset{\sim}{\psi}(\underset{\sim}{X}; \underset{\sim}{\theta})] = \underset{\sim}{0}$ AND (13.1)

$$E_{\underset{\sim}{\theta}}[\underset{\sim}{\psi}(\underset{\sim}{X}; \underset{\sim}{\theta})\, \underset{\sim}{\psi}(\underset{\sim}{X}; \underset{\sim}{\theta})^T] = I(\underset{\sim}{\theta}; \underset{\sim}{X}) \quad (13.2)$$

APPLY TAYLOR SERIES TO ESTIMATING EQUATIONS

(13.3) $$\underset{\sim}{0} = \sum_{i=1}^n \underset{\sim}{\psi}(\underset{\sim}{X}_i; \hat{\underset{\sim}{\theta}}_n) = \sum_{i=1}^n \underset{\sim}{\psi}(\underset{\sim}{X}_i; \underset{\sim}{\theta}) + H_n(\underset{\sim}{\theta})(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta})$$

$$+ \text{HIGHER TERMS}$$

where $H_n(\underset{\sim}{\theta}) = \sum_{i=1}^n H(\underset{\sim}{\theta}; \underset{\sim}{X}_i)$ with

$$H(\underset{\sim}{\theta}, \underset{\sim}{X}) = \underset{q\times q}{\begin{bmatrix} \nabla \psi_1(\underset{\sim}{X}; \underset{\sim}{\theta})^T \\ \vdots \\ \nabla \psi_2(\underset{\sim}{X}; \underset{\sim}{\theta})^T \end{bmatrix}} = \left[\frac{\partial^2 l(\underset{\sim}{X}; \underset{\sim}{\theta})}{\partial \underset{\sim}{\theta}_i \partial \underset{\sim}{\theta}_j} \right]_{i,j=1,\ldots,q} \quad \text{HESSIAN OF } l(\underset{\sim}{X}; \underset{\sim}{\theta})$$

HENCE $\frac{1}{n} H_n(\underset{\sim}{\theta}) \xrightarrow[\underset{\sim}{\theta}]{a.s.} E\left[H(\underset{\sim}{\theta}; \underset{\sim}{X})\right] = -I(\underset{\sim}{\theta}; \underset{\sim}{X})$

ALSO, ASSUME REGULARITY CONDITIONS SO THAT

$\frac{1}{\sqrt{n}}$ HIGHER TERMS $\xrightarrow[p]{} \underset{\sim}{0}$. (TAYLOR SERIES WITH REMAINDER TERMS)

SO, (13.3) $\Rightarrow$

$$-\left(\frac{1}{n} H_n(\underset{\sim}{\theta})\right) \sqrt{n}(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}) = \frac{1}{\sqrt{n}} \left(\sum_{i=1}^n \underset{q}{\psi}(\underset{\sim}{X}_i; \underset{\sim}{\theta}) \right) + o_p(1) \qquad (13.4)$$

BY (13.1) AND (13.2) AND MULTIVARIATE C.L.T.,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \underset{q}{\psi}(\underset{\sim}{X}_i; \underset{\sim}{\theta}) \xrightarrow{d} N_q(\underset{\sim}{0}, I(\underset{\sim}{\theta}; \underset{\sim}{X}))$$

$$\Rightarrow I(\underset{\sim}{\theta}; \underset{\sim}{X}) \sqrt{n}(\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}) \xrightarrow{d} N_q(\underset{\sim}{0}, I(\underset{\sim}{\theta}; \underset{\sim}{X})) \Rightarrow \text{RESULTS. } \square$$

SCORE'S TEST FOR $H_0: \underset{\sim}{\theta} = \underset{\sim}{\theta}_0$ BASED ON MLE

$$T_n = n(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0)' I(\underset{\sim}{\theta}_0; \underset{\sim}{X})(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0) \xrightarrow{d} \chi^2_q, \text{ SO}$$

ASYMPTOTIC α-LEVEL CRITICAL REGION: $T_n > \chi^2_{q,\alpha}$.

WALD'S TEST: REPLACE $I(\underset{\sim}{\theta}_0; \underset{\sim}{X})$ WITH $I(\hat{\underset{\sim}{\theta}}_n; \underset{\sim}{X})$

LIKELIHOOD RATIO STATISTIC FOR $H_0: \underset{\sim}{\theta} = \underset{\sim}{\theta}_0$

$$\Lambda_n = \frac{L(\underset{\sim}{\theta}_0; \underline{X})}{L(\hat{\underset{\sim}{\theta}}_n; \underline{X})} \qquad \text{REJECT WHEN TOO SMALL}$$

$$-2\log\Lambda_n = 2\left\{\sum_{i=1}^n l(\hat{\underset{\sim}{\theta}}_n; \underset{\sim}{X}_i) - \sum_{i=1}^n l(\underset{\sim}{\theta}_0; \underset{\sim}{X}_i)\right\}$$

NOW $\sum_{i=1}^n l(\hat{\underset{\sim}{\theta}}_n; \underset{\sim}{X}_i) = \sum_{i=1}^n l(\underset{\sim}{\theta}_0; \underset{\sim}{X}_i) + \sum_{i=1}^n \nabla l(\underset{\sim}{\theta}_0; \underset{\sim}{X}_i)^T (\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0)$

$$+ \frac{1}{2}(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0)^T H_n(\underset{\sim}{\theta}_0)(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0) + \text{HIGHER TERMS}$$

AND $\frac{1}{\sqrt{n}}\sum_{i=1}^n \nabla l(\underset{\sim}{\theta}_0; \underset{\sim}{X}_i) = \frac{1}{\sqrt{n}}\sum_{i=1}^n \underset{\sim}{\psi}(\underset{\sim}{X}_i; \underset{\sim}{\theta}_0) = -\left(\frac{1}{n}H_n(\underset{\sim}{\theta}_0)\right)\sqrt{n}(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0) + o_p(1)$

$\Rightarrow$ by (13.4)

(13.5) $$-2\log\Lambda_n = n(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0)^T\left(\frac{1}{n}H_n(\underset{\sim}{\theta}_0)\right)(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0) + o_p(1) \to \chi^2_q \text{ under } H_0$$

SINCE $\sqrt{n}(\hat{\underset{\sim}{\theta}}_n - \underset{\sim}{\theta}_0) \to N_q(\underset{\sim}{0}, \mathcal{I}_0 = I(\underset{\sim}{\theta}_0; X)^{-1})$

$\hat{\mathcal{I}}_0^{-1} = \frac{1}{n}H_n(\underset{\sim}{\theta}_0) \to I(\underset{\sim}{\theta}_0; \underset{\sim}{X}) = \mathcal{I}_0^{-1}$, AND SO

$n(\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}_0)'\hat{\mathcal{I}}_0^{-1}(\hat{\underset{\sim}{\theta}} - \underset{\sim}{\theta}_0) \to_d \chi^2_q$. HENCE,

THEOREM 13.4. UNDER $H_0: \underset{\sim}{\theta} = \underset{\sim}{\theta}_0$, $-2\log\Lambda_n \to_d \chi^2_q$.

COMPOSITE NULL HYPOTHESIS $H_0: \underset{\sim}{\theta} \in (H)_0 \subset (H)$

- SKETCH OF TREATMENT

$\Lambda_n = \dfrac{L(\hat{\underset{\sim}{\theta}}_0; \underline{X})}{L(\hat{\underset{\sim}{\theta}}; \underline{X})}$ where $\hat{\underset{\sim}{\theta}}$ IS MLE UNDER $\textcircled{H}$, AND

$\hat{\underset{\sim}{\theta}}_0$ IS MLE UNDER $\textcircled{H}_0$.

LIMITING DISTRIBUTION OF $\sqrt{n}(\hat{\underset{\sim}{\theta}} - \hat{\underset{\sim}{\theta}}_0)$ FOR $\textcircled{H}_0 = \textcircled{H} \cap V$ where

V is a hyperplane of dimension $p < q$.

CANONICAL FORM: $H_0 : \underset{\sim}{\theta} = \begin{pmatrix} \underset{\sim}{\theta}_1 \\ \underset{\sim}{0} \end{pmatrix}$, $\underset{\sim}{\theta}_1 \in \textcircled{H}_1 \subset \mathbb{R}^p$, $\text{VOL}[\textcircled{H}_1] > 0$.

LET $\hat{\underset{\sim}{\theta}} = \begin{pmatrix} \hat{\underset{\sim}{\theta}}_1 \\ \hat{\underset{\sim}{\theta}}_2 \end{pmatrix}$ and $\hat{\underset{\sim}{\theta}}_0 = \begin{pmatrix} \tilde{\underset{\sim}{\theta}}_1 \\ \underset{\sim}{0} \end{pmatrix}$. ALSO, $\underset{\sim}{\psi}(\underline{X}; \underset{\sim}{\theta}) = \begin{pmatrix} \underset{\sim}{\psi}_1(\underline{X}; \underset{\sim}{\theta}) \\ \underset{\sim}{\psi}_2(\underline{X}; \underset{\sim}{\theta}) \end{pmatrix}$.

AND $H(\underset{\sim}{\theta}; \underline{X}) = \begin{bmatrix} H_{11}(\underset{\sim}{\theta}; \underline{X}) & H_{12}(\underset{\sim}{\theta}; \underline{X}) \\ H_{21}(\underset{\sim}{\theta}; \underline{X}) & H_{22}(\underset{\sim}{\theta}; \underline{X}) \end{bmatrix}$, ANALOGOUS FOR $H_n(\underset{\sim}{\theta})$.

APPLYING (13.3) TO $\tilde{\underset{\sim}{\theta}}_1$ and let $\underset{\sim}{\theta}_0 = \begin{pmatrix} \underset{\sim}{\theta}_1 \\ \underset{\sim}{0} \end{pmatrix}$, then under H_0

$$-\left(\frac{1}{n} H_{n,11}(\hat{\underset{\sim}{\theta}}_0)\right) \sqrt{n}(\tilde{\underset{\sim}{\theta}}_1 - \underset{\sim}{\theta}_1) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \underset{\sim}{\psi}_1(\underline{X}_i; \underset{\sim}{\theta}_0) + o_p(1)$$

ALSO, FROM (15.4) WE HAVE UNDER H_0

$$-\left(\frac{1}{n} H_{n,11}(\underset{\sim}{\theta}_0)\right) \sqrt{n}(\hat{\underset{\sim}{\theta}}_1 - \underset{\sim}{\theta}_1) - \left(\frac{1}{n} H_{n,12}(\underset{\sim}{\theta}_0)\right) \sqrt{n}\, \hat{\underset{\sim}{\theta}}_2 = \frac{1}{n} \sum_{i=1}^n \underset{\sim}{\psi}_1(\underline{X}_i; \underset{\sim}{\theta}_0) + o_p(1)$$

$$\Rightarrow I_{11}(\underset{\sim}{\theta}_0) \sqrt{n}(\hat{\underset{\sim}{\theta}}_1 - \tilde{\underset{\sim}{\theta}}_1) - I_{12}(\underset{\sim}{\theta}_0) \sqrt{n}\, \hat{\underset{\sim}{\theta}}_2 \xrightarrow{p} \underset{\sim}{0}$$

HENCE, UNDER H_0

$$\sqrt{n}\left(\hat{\underset{\sim}{\theta}} - \hat{\underset{\sim}{\theta}}_0\right) = \begin{bmatrix} \sqrt{n}\left(\hat{\underset{\sim}{\theta}}_1 - \tilde{\underset{\sim}{\theta}}_1\right) \\ \sqrt{n}\,\hat{\underset{\sim}{\theta}}_2 \end{bmatrix} = \sqrt{n}\, A(\underset{\sim}{\theta}_0)\,\hat{\underset{\sim}{\theta}}_2 + o_p(1) \qquad (13.6)$$

where $\underset{q\times p}{A}(\underset{\sim}{\theta}) = \begin{bmatrix} I_{11}(\underset{\sim}{\theta})^{-1} I_{12}(\underset{\sim}{\theta}) \\ I_{q-p} \end{bmatrix}$

and so $\sqrt{n}\left(\hat{\underset{\sim}{\theta}} - \hat{\underset{\sim}{\theta}}_0\right) \to_d N_q\left(\underset{\sim}{0}, \mathcal{I}(\underset{\sim}{\theta}_0)\right)$ where

$$\mathcal{I}(\underset{\sim}{\theta}) = A(\underset{\sim}{\theta})\left\{I(\underset{\sim}{\theta})^{-1}\right\}_{22} A(\theta)^T \text{ with RANK } q-p.$$

$$\Rightarrow T_{0,n} = n\left(\hat{\underset{\sim}{\theta}} - \hat{\underset{\sim}{\theta}}_0\right)^T \mathcal{I}(\hat{\underset{\sim}{\theta}}_0)^{-}\left(\hat{\underset{\sim}{\theta}} - \hat{\underset{\sim}{\theta}}_0\right) \to \chi^2_{q-p} \quad \text{SCORE'S } \chi^2\text{-TEST}$$

BY (13.6), WE ALSO HAVE $T_{0,n} = n\,\hat{\underset{\sim}{\theta}}_2^T A(\hat{\underset{\sim}{\theta}}_0)^T \mathcal{I}(\hat{\underset{\sim}{\theta}}_0)^{-} A(\hat{\underset{\sim}{\theta}}_0)\,\hat{\underset{\sim}{\theta}}_2 + o_p(1)$

ALTERNATIVELY, SINCE $\sqrt{n}\left(\hat{\underset{\sim}{\theta}}_2 - \underset{\sim}{\theta}_2\right) \to_d N_{q-p}\left(\underset{\sim}{0}, \left\{I(\underset{\sim}{\theta}_0)^{-1}\right\}_{22}\right)$

IT FOLLOWS $n\,\hat{\underset{\sim}{\theta}}_2^T \left\{I(\hat{\underset{\sim}{\theta}}_0)^{-1}\right\}_{22}^{-1} \hat{\underset{\sim}{\theta}}_2 \to \chi^2_{q-p}$ under H_0

SHOW THE ALGEBRAIC IDENTITY

$$A(\hat{\underset{\sim}{\theta}}_0)^T \mathcal{I}(\hat{\underset{\sim}{\theta}}_0)^{-} A(\hat{\underset{\sim}{\theta}}_0) = \left\{I(\hat{\underset{\sim}{\theta}}_0)^{-1}\right\}_{22}^{-1} \qquad (13.7),$$

SIMILAR TO THE SIMPLE HYPOTHESIS CASE, IT CAN BE SHOWN

$$-2\log \Lambda_n = T_{0,n} + o_p(1) \quad \text{under } H_0$$

$$\to \chi^2_{q-p}.$$